

InferQ: A Database-Oriented Benchmark for Compositional Quantum Circuits: [Experiments & Analysis]

Abstract

We present InferQ, a comprehensive dataset of various combinational circuits and metrics / properties, designed to support quantum applications in the noisy intermediate-scale quantum (NISQ) era. InferQ defines a unified data model of features of structurally diverse circuits. It captures both quantum and classical properties, including graph-derived representations of circuit structure. We demonstrate its usefulness through three representative use cases: optimizing classical backend selection in Qiskit, extracting complex circuit features and profiling RDBMS runtimes for quantum. InferQ is publicly available and intended to be reused, reproduced and grown for quantum circuit analysis.

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References

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